

Lista 9

Exercícios de Ordenação

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Exercício 1

```
main.c
9   v[j] = x;
10  }
11
12  void bsort(int v[], int n) {
13      for(int i = 1; i < n; i++) {
14          for(int j = 0; j < n - i; j++) {
15              if(v[j] > v[j+1]) troca(v, j, j+1);
16          }
17      }
18  }
19
20  void exhibe(int v[], int n) {
21      for(int i = 0; i < n; i++) printf("%d ", v[i]);
22      printf("\n");
23  }
24
25  int main(void) {
26      int v[10] = {2, 16, 68, 60, 8, 65, 96, 2, 60, 42};
27      printf("Vetor original: ");
28      exhibe(v, 10);
29
30      bsort(v, 10);
31
input
Vetor original: 2 16 68 60 8 65 96 2 60 42
Vetor ordenado: 2 2 8 16 42 60 60 65 68 96
```

Exercício 2

```
main.c
1  #include <stdio.h>
2  /*
3   Marcos Vinícius Bispo de Oliveira | RA: 1840482513029
4   Ryan de Souza Silva | RA: 1840482513013
5  */
6  void troca(int v[], int i, int j) {
7      int x = v[i];
8      v[i] = v[j];
9      v[j] = x;
10 }
11
12 void empurra(int v[], int u) {
13     if (u == 0) return;
14     empurra(v, u - 1);
15     if (v[u - 1] > v[u]) {
16         troca(v, u - 1, u);
17     }
18 }
19
20 void exhibe(int v[], int n) {
21     printf("{");
22     for(int i = 0; i < n; i++) {
23         printf("%d", v[i], (i < n - 1) ? ", " : "");
input
Antes: {51,82,38,99,75,19,69,46,27}
Depois: {51,38,82,75,19,69,46,27,99}
```

Exercício 3

```
main.c
1  #include <stdio.h>
2  /*
3   Marcos Vinícius Bispo de Oliveira | RA: 18404825130
4   Ryan de Souza Silva | RA: 1840482513013
5  */
6  void troca(int v[], int i, int j) {
7      int x = v[i];
8      v[i] = v[j];
9      v[j] = x;
10 }
11
12 void empurra(int v[], int u) {
13     if (u == 0) return;
14     empurra(v, u - 1);
15     if (v[u - 1] > v[u]) troca(v, u - 1, u);
16 }
17
18 void bsr(int v[], int n) {
19     if (n <= 1) return;
20     empurra(v, n - 1);
21     bsr(v, n - 1);
22 }
23
input
Antes: {51,82,38,99,75,19,69,46,27}
Depois: {19,27,38,46,51,69,75,82,99}
```

Exercício 4

```
1  #include <stdio.h>
2  #include <stdlib.h>
3  /*
4   Marcos Vinícius Bispo de Oliveira | RA: 1840482513029
5   Ryan de Souza Silva | RA: 1840482513013
6  */
7  void intercala(int v[], int p, int m, int u) {
8      int *w = malloc((u - p + 1) * sizeof(int));
9      int i = p, j = m + 1, k = 0;
10
11     while(i <= m && j <= u) w[k++] = (v[i] < v[j]) ? v[i++] : v[j++];
12     while(i <= m) w[k++] = v[i++];
13     while(j <= u) w[k++] = v[j++];
14     for(k = 0; k <= u - p; k++) v[p + k] = w[k];
15     free(w);
16 }
17
18 void exhibe(int v[], int n) {
19     for(int i = 0; i < n; i++) printf("%d ", v[i]);
20     printf("\n");
21 }
22
23 int main(void) {
input
Vetor v antes: 31 48 60 80 19 27 52 75
Vetor v intercalado: 19 27 31 48 52 60 75 80

Vetor w antes: 10 82 27 38 41 53 60 75 99
Vetor w intercalado: 10 27 38 41 53 60 75 82 99
```

Exercício 5

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 /*
4  Marcos Vinícius Bispo de Oliveira / RA: 1840482513029
5  Ryan de Souza Silva / RA: 1840482513013
6 */
7 void intercala(int v[], int p, int m, int u) {
8     int *w = malloc((u - p + 1) * sizeof(int));
9     int i = p, j = m + 1, k = 0;
10    while(i <= m && j <= u) w[k++] = (v[i] < v[j]) ? v[i++] : v[j++];
11    while(i <= m) w[k++] = v[i++];
12    while(j <= u) w[k++] = v[j++];
13    for(k = 0; k <= u - p; k++) v[p + k] = w[k];
14    free(w);
15 }
16
17 void ms(int v[], int p, int u) {
18     if(p == u) return;
19     int m = (p + u) / 2;
20     ms(v, p, m);
21     ms(v, m + 1, u);
22     intercala(v, p, m, u);
23 }
```

input

Desordenado: 83 31 91 46 27 20 96 25 96 80
Ordenado: 20 25 27 31 46 80 83 91 96 96

Exercício 6

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 /*
4  Marcos Vinícius Bispo de Oliveira / RA: 1840482513029
5  Ryan de Souza Silva / RA: 1840482513013
6 */
7 void preenche(int v[], int n, int s) {
8     srand(s);
9     for(int i = 0; i < n; i++) v[i] = rand() % 1000;
10 }
11
12 void exhibe(int v[], int n) {
13     for(int i = 0; i < n; i++) printf("%d ", v[i]);
14     printf("\n");
15 }
16
17 int main(void) {
18     int v[15];
19     printf("Vetor preenchido com semente 1:\n");
20     preenche(v, 15, 1);
21     exhibe(v, 15);
22
23     printf("\nVetor preenchido com semente 42:\n");
24 }
```

input

Vetor preenchido com semente 1:
383 886 777 915 793 335 386 492 649 421 362 27 690 59 763

Vetor preenchido com semente 42:
166 740 881 241 12 758 21 940 535 743 874 143 717 504 696

Exercício 7

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <time.h>
4 /*
5  Marcos Vinicius Bispo de Oliveira / RA: 1840482513029
6  Ryan de Souza Silva / RA: 1840482513013
7 */
8 void troca(int v[], int i, int j) { int x = v[i]; v[i] = v
9
10 void bsort(int v[], int n) {
11     for(int i = 1; i < n; i++)
12         for(int j = 0; j < n - i; j++)
13             if(v[j] > v[j+1]) troca(v, j, j+1);
14 }
15
16 void intercala(int v[], int p, int m, int u) {
17     int *w = malloc((u - p + 1) * sizeof(int));
18     int i = p, j = m + 1, k = 0;
19     while(i <= m && j <= u) w[k++] = (v[i] < v[j]) ? v[i++]
20     while(i <= m) w[k++] = v[i++];
21     while(j <= u) w[k++] = v[j++];
22     for(k = 0; k <= u - p; k++) v[p + k] = w[k];
23     free(w);

```

input

Vetor preenchido com semente 1:
383 886 777 915 793 335 386 492 649 421 362 27 690 59 763

Vetor preenchido com semente 42:
166 740 881 241 12 758 21 940 535 743 874 143 717 504 696

Exercício 8

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <time.h>
4 /*
5  Marcos Vinicius Bispo de Oliveira / RA: 1840482513029
6  Ryan de Souza Silva / RA: 1840482513013
7 */
8 void intercala(int v[], int p, int m, int u) {
9     int *w = malloc((u - p + 1) * sizeof(int));
10    int i = p, j = m + 1, k = 0;
11    while(i <= m && j <= u) w[k++] = (v[i] < v[j]) ? v[i++]
12    while(i <= m) w[k++] = v[i++];
13    while(j <= u) w[k++] = v[j++];
14    for(k = 0; k <= u - p; k++) v[p + k] = w[k];
15    free(w);
16 }
17
18 void ms(int v[], int p, int u) {
19     if(p == u) return;
20     int m = (p + u) / 2;
21     ms(v, p, m); ms(v, m + 1, u); intercala(v, p, m, u);
22 }
23 void msort(int v[], int n) { ms(v, 0, n - 1); }

```

input

n	msort
1000000	0.18
2000000	0.36
3000000	0.55
4000000	0.75
5000000	0.95
6000000	1.14
7000000	1.37
8000000	1.53
9000000	1.78
10000000	1.98

Exercício 9

```
1 #include <stdio.h>
2 /*
3  Marcos Vinícius Bispo de Oliveira / RA: 1840482513029
4  Ryan de Souza Silva / RA: 1840482513013
5  */
6 int lsearch(int x, int v[], int n) {
7     for(int i = 0; i < n; i++) {
8         if(x == v[i]) return 1;
9     }
10    return 0;
11 }
12
13 int main(void) {
14     int v[8] = {66, 80, 31, 48, 27, 75, 19, 52};
15
16     printf("Procurando 27 (1 = Verdadeiro, 0 = Falso): %d\n", lsearch(27, v, 8));
17     printf("Procurando 51 (1 = Verdadeiro, 0 = Falso): %d\n", lsearch(51, v, 8));
18
19     return 0;
20 }
```

Exercício 10

```
1 #include <stdio.h>
2 /*
3  Marcos Vinícius Bispo de Oliveira / RA: 1840482513029
4  Ryan de Souza Silva / RA: 1840482513013
5  */
6 int busca_binaria(int x, int v[], int n) {
7     int p = 0;
8     int u = n - 1;
9     while(p <= u) {
10         int m = (p + u) / 2;
11         if(x == v[m]) return 1;
12         if(x < v[m]) u = m - 1;
13         else p = m + 1;
14     }
15     return 0;
16 }
17
18 int main(void) {
19     int v[8] = {19, 27, 31, 48, 52, 66, 75, 80};
20
21     printf("27: %d\n", busca_binaria(27, v, 8));
22     printf("51: %d\n", busca_binaria(51, v, 8));
23 }
```

input

```
27: 1
51: 0
```

Exercício 11

```
1 #include <stdio.h>
2 /*
3  Marcos Vinícius Bispo de Oliveira / RA: 1840482513029
4  Ryan de Souza Silva / RA: 1840482513013
5  */
6 int rlsearch(int x, int v[], int n) {
7
8     if (n == 0) return 0;
9
10
11     if (v[n - 1] == x) return 1;
12
13
14     return rlsearch(x, v, n - 1);
15 }
16
17 int main(void) {
18     int v[8] = {66, 80, 31, 48, 27, 75, 19, 52};
19
20     printf("Procurando 75: %d\n", rlsearch(75, v, 8));
21     printf("Procurando 10: %d\n", rlsearch(10, v, 8));
22
23     return 0;
24 }
```

input

Procurando 75: 1
Procurando 10: 0

Exercício 12

```
1 #include <stdio.h>
2 /*
3  Marcos Vinícius Bispo de Oliveira / RA: 1840482513029
4  Ryan de Souza Silva / RA: 1840482513013
5  */
6 int rbsearch(int x, int v[], int p, int u) {
7
8     if (p > u) return 0;
9
10     int m = (p + u) / 2;
11
12
13     if (x == v[m]) return 1;
14
15
16     if (x < v[m]) return rbsearch(x, v, p, m - 1);
17
18
19     return rbsearch(x, v, m + 1, u);
20 }
21
22 int main(void) {
23
24 }
```

input

Procurando 80: 1
Procurando 15: 0